Power Question 2011: Power of Triangles

The arrangement of numbers known as Pascal's Triangle has fascinated mathematicians for centuries. In fact, about 700 years before Pascal, the Indian mathematician Halayudha wrote about it in his commentaries to a then-1000-year-old treatise on verse structure by the Indian poet and mathematician Pingala, who called it the *Meruprastāra*, or "Mountain of Gems". In this Power Question, we'll explore some properties of Pingala's/Pascal's Triangle ("PT") and its variants.

Unless otherwise specified, the **only** definition, notation, and formulas you may use for PT are the definition, notation, and formulas given below.

PT consists of an infinite number of rows, numbered from 0 onwards. The nth row contains $n+1$ numbers, identified as $\text{Pa}(n,k)$, where $0 \le k \le n$. For all n, define $\text{Pa}(n,0) = \text{Pa}(n,n) = 1$. Then for $n > 1$ and $1 \le k \le n-1$, define $\text{Pa}(n,k) = \text{Pa}(n-1,k-1) + \text{Pa}(n-1,k)$. It is convenient to define $\text{Pa}(n,k) = 0$ when $k < 0$ or $k > n$. We write the nonzero values of PT in the familiar pyramid shown below.

			1			
			$\text{Pa}(0,0)$			
		1		**1**		
		$\text{Pa}(1,0)$		$\text{Pa}(1,1)$		
	1		**2**		**1**	
	$\text{Pa}(2,0)$		$\text{Pa}(2,1)$		$\text{Pa}(2,2)$	
1		**3**		**3**		**1**
$\text{Pa}(3,0)$		$\text{Pa}(3,1)$		$\text{Pa}(3,2)$		$\text{Pa}(3,3)$

As is well known, $\text{Pa}(n,k)$ gives the number of ways of choosing a committee of k people from a set of n people, so a simple formula for $\text{Pa}(n,k)$ is $\text{Pa}(n,k) = \frac{n!}{k!(n-k)!}$. You may use this formula or the recursive definition above throughout this Power Question.

1a. For $n = 1, 2, 3, 4$, and $k = 4$, find $\text{Pa}(n,n) + \text{Pa}(n+1,n) + \cdots + \text{Pa}(n+k,n)$. [2]

1b. If $\text{Pa}(n,n) + \text{Pa}(n+1,n) + \cdots + \text{Pa}(n+k,n) = \text{Pa}(m,j)$, find and justify formulas for m and j in terms of n and k. [2]

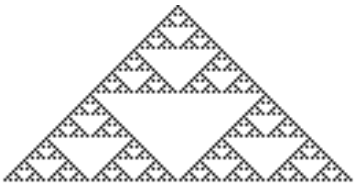
Consider the *parity* of each entry: define

$$\text{PaP}(n,k) = \begin{cases} 1 & \text{if } \text{Pa}(n,k) \text{ is odd,} \\ 0 & \text{if } \text{Pa}(n,k) \text{ is even.} \end{cases}$$

2a. Prove that $\text{PaP}(n,0) = \text{PaP}(n,n) = 1$ for all nonnegative integers n. [1]

2b. Compute rows $n = 0$ to $n = 8$ of PaP. [2]

You may have learned that the array of parities of Pascal's Triangle forms the famous fractal known as Sierpinski's Triangle. The first 128 rows are shown below, with a dot marking each entry where $\text{PaP}(n,k) = 1$.



The next problem helps explain this surprising connection.

3a. If $n = 2^j$ for some nonnegative integer j, and $0 < k < n$, show that $\mathrm{PaP}(n, k) = 0$. [2]

3b. Let $j \geq 0$, and suppose $n \geq 2^j$. Prove that $\mathrm{Pa}(n, k)$ has the same parity as the sum $\mathrm{Pa}(n - 2^j, k - 2^j) + \mathrm{Pa}(n - 2^j, k)$, i.e., either both $\mathrm{Pa}(n, k)$ and the given sum are even, or both are odd. [2]

3c. If j is an integer such that $2^j \leq n < 2^{j+1}$, and $k < 2^j$, prove that

$$\mathrm{PaP}(n, k) = \mathrm{PaP}(n - 2^j, k). \quad [2]$$

Clark's Triangle: If the left side of PT is replaced with consecutive multiples of 6, starting with 0, but the right entries (except the first) and the generating rule are left unchanged, the result is called *Clark's Triangle*. If the kth entry of the nth row is denoted by $\mathrm{Cl}(n, k)$, then the formal rule is:

$$\begin{cases} \mathrm{Cl}(n, 0) = 6n & \text{for all } n, \\ \mathrm{Cl}(n, n) = 1 & \text{for } n \geq 1, \\ \mathrm{Cl}(n, k) = \mathrm{Cl}(n-1, k-1) + \mathrm{Cl}(n-1, k) & \text{for } n \geq 1 \text{ and } 1 \leq k \leq n-1. \end{cases}$$

The first four rows of Clark's Triangle are given below.

			0			
			$\mathrm{Cl}(0,0)$			
		6		**1**		
		$\mathrm{Cl}(1,0)$		$\mathrm{Cl}(1,1)$		
	12		**7**		**1**	
	$\mathrm{Cl}(2,0)$		$\mathrm{Cl}(2,1)$		$\mathrm{Cl}(2,2)$	
18		**19**		**8**		**1**
$\mathrm{Cl}(3,0)$		$\mathrm{Cl}(3,1)$		$\mathrm{Cl}(3,2)$		$\mathrm{Cl}(3,3)$

4a. Compute the next three rows of Clark's Triangle. [2]

4b. If $\mathrm{Cl}(n, 1) = an^2 + bn + c$, determine the values of a, b, and c. [2]

4c. Prove the formula you found in 4b. [2]

5a. Compute $\mathrm{Cl}(11, 2)$. [1]

5b. Find and justify a formula for $\mathrm{Cl}(n, 2)$ in terms of n. [2]

5c. Compute $\mathrm{Cl}(11, 3)$. [1]

5d. Find and justify a formula for $\mathrm{Cl}(n, 3)$ in terms of n. [2]

6. Find and prove a closed formula (that is, a formula with a fixed number of terms and no "…") for $\mathrm{Cl}(n, k)$ in terms of n, k, and the Pa function. [3]

Leibniz's Harmonic Triangle: Consider the triangle formed by the rule

$$\begin{cases} \mathrm{Le}(n, 0) = \frac{1}{n+1} & \text{for all } n, \\ \mathrm{Le}(n, n) = \frac{1}{n+1} & \text{for all } n, \\ \mathrm{Le}(n, k) = \mathrm{Le}(n+1, k) + \mathrm{Le}(n+1, k+1) & \text{for all } n \text{ and } 0 \le k \le n. \end{cases}$$

This triangle, discovered first by Leibniz, consists of reciprocals of integers as shown below.

$$\begin{array}{ccccccc} & & & \mathbf{1} & & & \\ & & & \mathrm{Le}(0,0) & & & \\ & & \mathbf{\frac{1}{2}} & & \mathbf{\frac{1}{2}} & & \\ & & \mathrm{Le}(1,0) & & \mathrm{Le}(1,1) & & \\ & \mathbf{\frac{1}{3}} & & \mathbf{\frac{1}{6}} & & \mathbf{\frac{1}{3}} & \\ & \mathrm{Le}(2,0) & & \mathrm{Le}(2,1) & & \mathrm{Le}(2,2) & \\ \mathbf{\frac{1}{4}} & & \mathbf{\frac{1}{12}} & & \mathbf{\frac{1}{12}} & & \mathbf{\frac{1}{4}} \\ \mathrm{Le}(3,0) & & \mathrm{Le}(3,1) & & \mathrm{Le}(3,2) & & \mathrm{Le}(3,3) \end{array}$$

For this contest, you may assume that $\mathrm{Le}(n, k) > 0$ whenever $0 \le k \le n$, and that $\mathrm{Le}(n, k)$ is undefined if $k < 0$ or $k > n$.

7a. Compute the entries in the next two rows of Leibniz's Triangle. [1]

7b. Compute $\mathrm{Le}(17, 1)$. [1]

7c. Compute $\mathrm{Le}(17, 2)$. [1]

8a. Find and justify a formula for $\mathrm{Le}(n, 1)$ in terms of n. [2]

8b. Compute $\displaystyle\sum_{n=1}^{2011} \mathrm{Le}(n, 1)$. [2]

8c. Find and justify a formula for $\mathrm{Le}(n, 2)$ in terms of n. [2]

9a. If $\displaystyle\sum_{i=1}^{\infty} \mathrm{Le}(i, 1) = \mathrm{Le}(n, k)$, determine the values of n and k. [2]

9b. If $\sum_{i=m}^{\infty} \mathrm{Le}(i, m) = \mathrm{Le}(n, k)$, compute expressions for n and k in terms of m. [2]

9c. Justify your result in 9b. [2]

10. Find three distinct sets of positive integers $\{a, b, c, d\}$ with $a < b < c < d$ such that

$$\frac{1}{3} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}\,. \qquad [3]$$

11. Find and prove a closed formula (that is, a formula with a fixed number of terms and no "...") for $\mathrm{Le}(n, k)$ in terms of n, k, and the Pa function. [4]